Probability Part 3

MATHEMATICAL EXPECTATION

A **mathematical expectation** is the average amount a player can expect to win or lose on one play in any game of chance.

It can be found by multiplying the probability of each possible outcomes by its payoff, and then adding these results.

If the expectation is positive, the player will win in the long run and lose if the expectation is negative.

Problem:
In tossing a single coin, P 100 bet is placed on heads i.e. if heads comes up, the player wins P 100 and if tails comes up, the player loses P 100. What is the expectation?

Solution:

	Heads	Tails
Probability	½	½
Payoff	100	-100

$$\text{Expectation} = \left(\frac{1}{2}\right)(100) + \left(\frac{1}{2}\right)(-100)$$

$$\text{Expectation} = 0$$

Problem:
In Casino Filipino, a roulette offers winning a bet on a single number pays 35 to 1. The actual probability is 1 in 38. What is the expectation?

Solution:

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The 35 to 1 is a house (payoff) odds of Casino Filipino.

Expectation can be obtained by taking the difference between the product of the first number in the house odds and probability of success, and the product of the second number of the house odds and the probability of failure.

$$\text{Expectation} = 35\left(\frac{1}{38}\right) - 1\left(\frac{37}{38}\right)$$

$$\text{Expectation} = -\frac{2}{38}$$

$$\text{Expectation} = -0.0526$$

Thus, a player expects to lose to Casino Filipino an average of 5.26 centavos for every peso bet.

Card Games

Card games are games which are played using a standard deck of cards. A standard deck of cards contains 52 cards divided into 4 suits; spades, hearts, diamonds and clubs. Each suit contains 13 cards labeled Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, jack, queen and king. Each of the four kings in a deck represents a great leader from history; Charlemagne (hearts), Alexander the Great (clubs), Julius Ceasar (diamonds), and King David (spades).

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The playing cards were used also to describe a calendar year. The 52 cards represent the 52 weeks in a year. The 4 suits represent the four seasons of the year and the 12 face cards (4 kings, 4 queens and 4 jacks) represent the 12 months of the year.

Poker Hand with corresponding number of ways:

Hand	No. of ways
Royal flush	4
Straight flush	36
Four of a kind	624
Full house	3744
Flush	5108
Straight	10200
Three of a kind	54912
Two pair	123552
Pair	1098240
None of the above	1302540
Total	2598960

Poker Hand with corresponding probability:

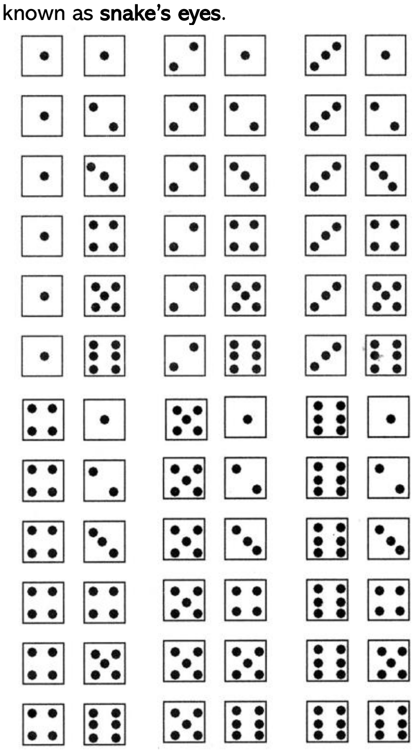
Hand	Probability
Royal flush	0.00000154
Straight flush	0.00001385
Four of a kind	0.0002401
Full house	0.0014406
Flush	0.0019654
Straight	0.0039246
Three of a kind	0.0211285
Two pair	0.0475390
Pair	0.4225690
None of the above	0.5011774
Total	1.00000000

Poker Hand corresponding expected frequency:

Hand	Expected frequency
Royal flush	1 in 649740 hands
Straight flush	1 in 72192 hands
Four of a kind	1 in 4165 hands
Full house	1 in 694 hands
Flush	1 in 509 hands
Straight	1 in 255 hands
Three of a kind	1 in 47.33 hands
Two pair	1 in 21 hands
Pair	1 in 2.37 hands
None of the above	1 in 2 hands

Probability with Dice

Dice were invented for sole purpose of gambling. As a matter of fact, dice games have so little intrinsic interest that in the absence of wagering they would hardly be worth playing. Dice were first used by the Chinese. The sum of the opposite faces of a die is always equal to 7. And the sum of all the vertical faces of a die, no matter how it rolls is always equal to 14. When two dice are rolled and both will face up 1, this is known as **snake's eyes**.



The following are the two-dice probabilities:

Two-dice sum	Probability
2	1/36
3	2/36
4	3/36
5	4/36
6	5/36
7	6/36
8	5/36
9	4/36
10	3/36
11	2/36
12	1/36

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